GU Tech - Al Ghazali University

Midterm Exam Spring 2025

Department of Computer Science

CS101 - Problem Solving & Programming Fundamentals (Lab)

02 July 2025 9:00 am - 12:00 pm Total marks: 30

Instructions

- Attempt all questions. There are 3 questions for a total of 30 marks.
- This is an open book exam but peer discussions and Internet access are not allowed. You may bring and use your laptops and lab manuals.
- Download the starter code from this GitHub repository. Be done within 5 minutes of being seated; internet access will be turned off thereafter.
  - You may use any indentation style but be consistent.
  - Do **not** modify or remove:
    - the function prototypes.
    - the print statements within the functions.
    - the `if (argc > 1)` block in the `main` function.
  - The starter code is also provided at the end of this paper. If you are late and unable to download the code from GitHub, you may copy from here. Be advised that copying from a PDF will ruin code formatting.
- Write clean code. Dirty hacks are grounds for marks deduction. They include but are not limited to premature `return` statements in `void` functions, unnecessarily breaking out of `for` loops, etc.
- Submit your code at both Google and GitHub classrooms.

Questions

1. The natural logarithm can be approximated by the following series: (10)

$$\frac{x-1}{x} + \frac{1}{2}\left(\frac{x-1}{x}\right)^2 + \frac{1}{2}\left(\frac{x-1}{x}\right)^3 + \frac{1}{2}\left(\frac{x-1}{x}\right)^4 + \cdots$$

Complete the function to calculate the sum of the first seven terms of this series.

`double naturalLog(double x)`

Hints:

- You should complete and make use of the helper function `double power(double val, int pow)`.
- Use `double` literals instead of integers where floating point calculations are necessary. For example, instead of using `1 / 2`, use either `0.5` or `1.0 / 2.0`.
  - That is because `1 / 2` is integer division and will return **0** instead of **0.5**.

The following table provides some sample input and output values.

Input `scanf("%lf", &x);`	**Output** `printf("%f\n", naturalLog(x))`
1	0.000000
1.5	0.416552
2.0	0.746094
2.71828	1.140553

2. Complete the function `void secondsToHoursMinutesSeconds(uint64_t s)` that converts given number of seconds to hours, minutes and seconds and prints them in the HH:MM:SS format. Test cases include: (10)

Input `uint32_t s`	**Output** `printf("%02d:%02d:%02d\n", hours, minutes, seconds)`
10	00:00:10
60	00:01:00
70	00:01:10
120	00:02:00
150	00:02:30
300	00:05:00
330	00:05:30
3600	01:00:00
3630	01:00:30
3670	01:01:10
7340	02:02:20
8274	02:17:54

3. **Parking Garage Revenue Calculator:** You are managing a parking garage with f floors, each containing s parking slots. (10)

   Each slot can be empty or occupied. If occupied, the parking fee is determined based on the following rules:

   - Vehicles parked for ≤ 3 hours: Rs. 10/hour
   - Vehicles parked for > 3 hours: Rs. 8/hour for each **additional** hour

   Write a program that calculates the total revenue generated for the entire garage.

   Input:

   - f (number of floors)
   - s number of slots per floor
   - Parking hours for each slot (enter 0 if empty)

   Output: Total revenue

   Note: There is no starter code for this question.

Starter Code

Question 1

```c
#include <stdio.h>

double power(double val, int pow);
double naturalLog(double x);

int main()
{
        double x = 0;

        printf("Enter x: ");
        scanf("%lf", &x);
//        printf("%f\n", power(2, 5));
        printf("%f\n", naturalLog(x));

        return 0;
}

double power(double val, int pow)
{
        //Student code goes here.

        return 0;
}

double naturalLog(double x)
{
        //Student code goes here.

        return 0;
}
```

Question 2

```c
#include <stdio.h>

void secondsToHoursMinutesSeconds(int s);

int main()
{
        int s;
        printf("Enter seconds: ");
        scanf("%d", &s);
        secondsToHoursMinutesSeconds(s);

        return 0;
}

void secondsToHoursMinutesSeconds(int s)
{
        int hours = 0, minutes = 0, seconds = 0;

        //Student code goes here.

        printf("%02d:%02d:%02d\n", hours, minutes, seconds);
}
```